

Fig. 9: An actual-size PCB pattern for XBee-controlled aircraft (receiver side)

D1, D2, D3 as 2 (as analogue data entry). However, these pins can be made digital data input or output pins also by simply changing to 3, 4 or 5. Set D5 to 4. This pin will be used as feedback signal from receiver XBee (XB2).

Now come to I/O Sampling section and set the IR sampling to 64, that is, 0X64. Once it is on 100-millisecond the feedback will be checked.

Check Write and let the configuration be written in XBee (XB2).

Construction and testing

An actual size, single-side PCB for the XBee-controlled aircraft transmitter circuit is shown in Fig. 7 its component layouts in Fig. 8. PCB for receiver circuit is shown in Fig. 9 and its component layout in Fig. 10.

PCB of receiver board1 is in the form of shield, so mount it on top of Arduino UNO board. Connect 3.7V

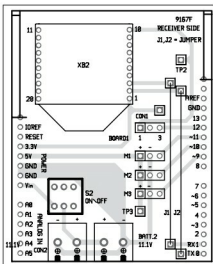


Fig. 10: Component layout for the PCB (receiver side)

battery to transmitter section and 11.1V battery to receiver section. Check working of BLDC motor by varying VR1, of servo motor M1 using VR4, of servo motor M2 using VR2 and of servo motor M3 using VR3. If there is any problem, verify voltages at test points given in the table.

Building an RC aircraft requires some basic skills and creativity. There are rich tutorials available on the Internet. Some useful information to build your own RC aircraft can be found from the links below:

<http://www.easycrc.com/airplanes/>

<http://rcvehicles.about.com/od/diyaircraft/>

<http://www.stenulson.net/rc-flight/rcflight.htm>

You can either build the body of the aircraft yourself or get it ready-made from an electronics spares shop.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

The receiver PCB, along with the 11.1V battery, should be mounted firmly in the aircraft.

The RC aircraft can be launched by gathering speed on a long runway while some are launched by hand by literally throwing them into the air. In the first case, as soon as your plane builds up enough speed and catches wind, it will lift from the ground. But note that, if the runway surface is uneven (like grass-covered), your plane might not be able to gather the necessary speed for take-off. In that case you should opt for hand-launch.

Remember, the aircraft is controlled by a remote control which has a finite range. If your plane gets too far away from you (either vertically or horizontally), you might lose control and the plane would plummet back to the ground. The usual way to fly it is in circles above your head.

After assembling the complete project, switch on transmitter and receiver circuits. Slowly vary VR1 to increase the speed of the BLDC motor (propeller). Once the plane takes off and is stabilised above the ground, vary VR2, VR3 and VR4 one by one, gradually, and observe the effect on flight pattern. To return plane back to ground, reduce its speed using VR1. ●

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